HSLU Hochschule Luzern

facADe: A Data-Driven Framework for City-Wide Facade Advertising Visibility and Monetization

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GitHub: https://github.com/fcampell/DWDL

Facade Impressions (Dashboard 1):
https://public.tableau.com/app/profile/leonard.dost/viz/FacADeDashbaord/FacADeDashboard
Facade Revenue (Dashboard 2):
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1. PROJECT IDEA AND USE CASE

1.1. Topic Introduction

The digitalization of physical advertising spaces has significantly changed how urban environments are evaluated for commercial visibility. In Switzerland, outdoor advertising is traditionally dominated by a small number of established out-of-home (OOH) providers, most notably APG|SGA, which operates analogue and digital advertising infrastructure across streets, transport hubs, and other high-mobility public spaces [1] , [2] . Planning decisions in this domain are typically supported by manual scouting, historical booking data, and standardized performance metrics derived from aggregated mobility studies [3] .

While these approaches are effective for premium, centrally managed advertising locations, they rarely provide a city-wide, data-driven view of *all* potentially viable advertising surfaces. In particular, building facades that are not part of established advertising networks remain largely unassessed, despite being exposed to substantial pedestrian, cyclist, and vehicular flows. At the same time, recent market analyses highlight that the value of OOH and DOOH advertising increasingly depends on the integration of mobility data and contextual signals to better quantify reach and return on investment [4] .

Modern marketing therefore operates at the intersection of urban mobility, geospatial analytics, and real estate utilization. Agencies require granular, time-resolved insights into how people move through cities, while property owners seek transparent mechanisms to evaluate whether facade advertising is economically attractive. This creates a multi-stakeholder ecosystem, summarized in Table 1, where demand for visibility and supply of physical space are closely interlinked.

Table 1: Stakeholders in Switzerland's Physical Advertisement Ecosystem

Stakeholder Type	Examples	Role in Ecosystem
Marketing Agencies	Clear Channel, APG	Plan and sell advertising visibility
Advertisers (Brands & Companies)	Retail chains, insurance companies, telecom operators	Purchase advertising visibility to reach target audiences
Real Estate Owners / Portfolio Managers	Private landlords, institutional investors, pension funds, real-estate companies	Own facades and physical surfaces; may rent facade space for advertising
Municipalities & Public Authorities	City planning departments, transport authorities	Regulate urban space usage and issue permits for outdoor advertising
Technology & Data Providers	Mobility data services, geospatial analytics providers	Provide geodata, mobility streams, and analytical tools for visibility modelling

Traditional outdoor advertising markets are characterized by scarcity: a limited number of premium surfaces command high prices. The core idea of this project is to expand this market by identifying previously underutilized facade surfaces that offer meaningful visibility but are cheaper and easier to deploy. By systematically quantifying the visibility potential of all building facades in the city, the competitive focus shifts from a small set of premium locations to a broader, data-driven supply of advertising opportunities.

1.2. Use Cases: Demand and Supply Perspectives

The project addresses two complementary use cases that together form a two-sided market. From the **demand-side**, marketing agencies require a reliable way to identify facades with high advertising potential. Beyond total traffic volumes, agencies are interested in how exposure varies by mobility mode (pedestrians, cyclists, motorized traffic, and public transport) as well as by time of day. Pedestrian-dominated exposure is particularly relevant, as slower-moving traffic typically yields longer viewing durations and higher recall rates [5] [6] . The proposed solution enables agencies to search, filter, and rank facades based on multi-modal visibility metrics, thereby expanding the inventory of viable advertising locations and supporting more flexible and cost-efficient campaign planning.

From the **supply-side**, real estate owners and portfolio managers face uncertainty about whether facade advertising is economically worthwhile. Even when facades are physically suitable, owners often lack transparent estimates of potential revenues and may perceive the process as operationally complex. The platform therefore provides a dedicated dashboard that allows owners to assess individual buildings or entire portfolios based on visibility-derived impression counts and benchmarked pricing assumptions. This creates a data-driven basis for evaluating monetization potential and comparing assets within a portfolio.

Both use cases converge within a shared analytical foundation. By anchoring analyses at facade level and enabling aggregation to buildings and portfolios using EGID (*Eidgenössische Gebäude ID*) as a common identifier, the system ensures consistency across stakeholder perspectives. Over time, the integration of demand-side usage data and realized prices can further improve transparency and pricing accuracy within the market.

1.3. Motivation

The motivation for this project arises from the complementary expertise within the project team. Practical experience from the advertising sector provides insight into campaign pricing, operational constraints, and agency decision-making processes. This is complemented by experience from the real estate domain, including asset valuation, portfolio management, and revenue optimization. The intersection of these domains forms a strong foundation for developing a technically robust solution that addresses concrete business needs on both the demand and supply side.

1.4. Problem Definition and Goals

Despite the availability of open mobility and geospatial data, no comprehensive system currently exists that consistently evaluates facade advertising potential across an entire city. Existing approaches rely on partial datasets, isolated traffic indicators, or manual assessments that do not scale and fail to capture multi-modal mobility dynamics.

The primary goal of this project is therefore to design and implement an end-to-end, data-driven system that quantifies facade visibility using real mobility patterns and makes the results accessible through analytical dashboards. This includes automated data ingestion, city-wide mobility modelling, geospatial visibility analysis, and a data warehouse that supports both operational and strategic decision-making.

1.5. Business and Analytical Questions

At the data lake and modelling level, the project addresses core analytical questions: which facades generate the highest number of potential visual impressions, how these impressions are distributed across mobility modes and time of day, and how facade visibility varies spatially across the city. These questions guide the integration of mobility flows and facade geometries into a unified visibility model.

At the data warehouse and business intelligence level, the focus shifts to strategic questions relevant for decision-makers. These include identifying high-potential buildings within a portfolio, comparing advertising value across districts, and deriving key performance indicators for both marketing agencies and real estate owners. Together, these questions ensure that the technical implementation remains tightly aligned with business objectives and evaluation criteria.

2. DATA LAKE

2.1. Data Sources

Our data lake integrates complementary datasets that enable a city-scale model of facade advertising visibility. Zürich Open Government Data (OGD) provides the main mobility inputs, which we augment with our own modelling pipelines to create city-wide, time-resolved flow estimates and facade visibility outputs. Figure 1 summarizes the end-to-end data flow from ingestion to transformation and model production within the lake.

To estimate daily visibility for every facade in Zürich, we require (1) detailed **facade geometries** and (2) **mobility and flow datasets**. Facade geometries define the physical advertising surfaces and are the spatial basis for all subsequent calculations. Mobility datasets describe how people traverse the city across different modes; motorized traffic, pedestrians and bicycles, and public transport and therefore determine how often a facade is potentially seen throughout the day. Combining static geospatial data with dynamic mobility data directly supports our business questions: identifying facades with the highest potential impressions and enabling downstream monetization and valuation use cases.

Table 2: Data Sources

Dataset	***Type***	***Description***	***Update Frequency***	***Access Method***	***Download***
Facade Geometries *(GeoCat)*	Static	High-resolution 2D facade geometries for all buildings in Zürich, derived from the "Solar energy: suitability of facade" dataset	Irregular	Direct file download (GeoPackage)	https://opendata.swiss/en/dataset/eignung-von-hausfassaden-fur-die-nutzung-von-sonnenenergie
Motorized Traffic Counts (MIV)	Dynamic	Hourly vehicle counts from measurement stations across Zürich	Daily	CKAN API	https://data.stadt-zuerich.ch/dataset/sid_dav_verkehrszaehlung_miv_od2031
Pedestrian & Bicycle Counts	Dynamic	Hourly pedestrian and bicycle counts from fixed counters	Daily	CKAN API	https://data.stadt-zuerich.ch/dataset/ted_taz_verkehrszaehlungen_werte_fussgaenger_velo
VBZ Passenger Volumes	Dynamic	Passenger counts per day type and per line segment	Monthly	Direct file download via OGD portal	https://data.stadt-zuerich.ch/dataset/vbz_fahrgastzahlen_ogd
OpenStreetMap	Dynamic	Street and path network used as topological reference for propagating traffic, pedestrian, and bicycle flows onto a continuous city-wide graph	Daily	Fetched within containerized flow-propagation models	https://www.openstreetmap.org/#map=8/46.825/8.224

In addition to the primary data sources listed, our propagation models rely on the current street and path network derived from OpenStreetMap (OSM). This network is

not ingested into the data lake as a persistent dataset but is fetched on demand within the modelling containers and used as a topological reference for flow propagation.
While the City of Zürich publishes extensive mobility datasets, they are not uniformly distributed or complete across all urban areas. Traffic and pedestrian/bicycle measurements are collected at selected counting stations; peripheral regions have fewer stations, requiring *propagation modelling* to estimate flows beyond measurement points. VBZ passenger data provides counts per segment or stop but does not include geometries, so georeferencing requires additional matching to stop coordinates and line shapes. Facade geometries are static and highly detailed, but updates may lag behind renovations or newly constructed buildings. These limitations justify the medallion architecture shown in *Figure 1*: raw datasets land in Bronze, are standardized in Silver, and are modelled in Gold, where propagation and visibility models address spatial coverage gaps and produce comprehensive, city-wide outputs.

Data Catalog, Governance, and Metadata
All datasets (raw, cleaned, and modelled) are registered in the AWS Glue Data Catalog. For each dataset we store at least the name, database/location, classification, and deprecation status. This metadata layer ensures that downstream consumers (ECS jobs, notebooks, and BI dashboards) access consistent and governed datasets.

Schema Strategy, Versioning, and Lineage
Mobility datasets evolve over time; fields may appear, disappear, or change formats. To handle this, our data lake follows a clear separation of responsibilities: Bronze is schema-on-read and stores raw files exactly as provided; Silver enforces standardized timestamps, geometry fields, ID columns, and naming conventions; and Gold stores versioned model outputs. Each modelling pipeline (facade visibility, motorized flow propagation, pedestrian/bike propagation, and VBZ integration) writes version-tagged outputs that document the model version and the input dataset. *Table 2* lists our sources, update type, and access method. New data is fetched weekly and ingested into the lake.
A central limitation across the dynamic mobility sources is that pedestrian, bicycle, and motorized traffic measurements originate from counting stations. Deriving full-coverage flows requires propagation models, which introduces uncertainty that must be considered in downstream analytics.

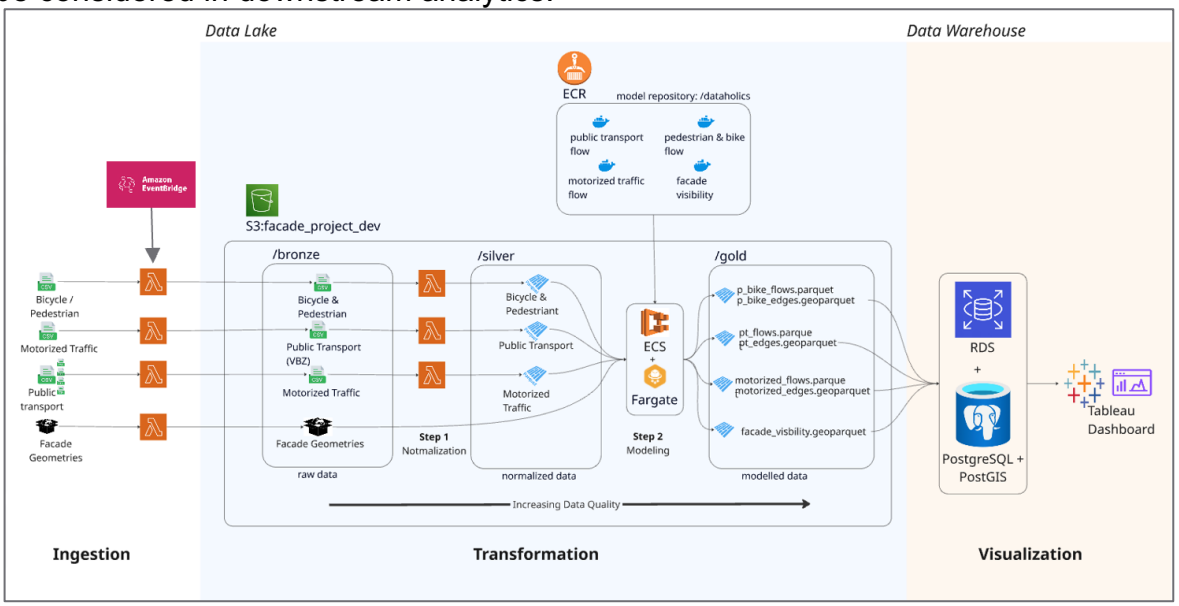


Figure 1: End-to-End Data Lake and Warehouse Architecture

2.2. Architecture

The data lake architecture follows a **medallion structure** with Bronze, Silver, and Gold layers that organize data by quality, usability, and readiness for analysis. Figure 1 illustrates the end-to-end architecture from extraction to dashboard consumption.
The solution is deployed on AWS, using Amazon S3 as the core storage layer, AWS Glue Data Catalog for metadata management, AWS Lambda for ingestion orchestration, and AWS ECS Fargate for executing geospatial modelling pipelines. Amazon RDS provides the data warehouse layer used by the dashboard. This modular setup allows independent scaling of ingestion, modelling, and analytics components.

Table 3: Medallion Data Lake structure

Layer	***Role***	***Description***
Bronze	Raw ingestion	CSVs and Parquets fetched via Lambda from Zürich OGD. No changes applied.
Silver	Standardized data	Scrubbed, normalized, enriched (consistent timestamps, harmonized schemas, geometry conversion).
Gold	Business-ready outputs	Final facade visibility tables, propagated flows, and curated datasets prepared for BI dashboards and real estate evaluation.

Storage Layers & Zones
Table 3 summarizes the role of each medallion layer. Bronze stores raw ingested data without modifications, Silver stores standardized datasets (consistent timestamps, harmonized schemas, geometry conversions), and Gold stores business-ready outputs, including final facade visibility tables and propagated flow datasets prepared for BI consumption and real-estate evaluation. Gold therefore acts as the authoritative foundation for both use cases: marketing agencies (visibility search) and property owners (revenue estimation).

File Formats
We use Parquet for all tabular datasets, GeoParquet and GeoPackage for spatial layers, and CSV only at ingestion when a source provides CSV. This combination supports efficient analytical processing while maintaining GIS interoperability where needed.

Compute Engines
Each modelling step is executed as an independent ECS Fargate container (facade visibility, motorized flow propagation, pedestrian/bike flow modelling, and VBZ passenger modelling). Within these containers, the OpenStreetMap network is retrieved dynamically and processed using NetworkX (python library) and OSM-based utilities to perform graph-based flow propagation. This design improves reproducibility and isolation of dependencies, enables independent scaling, and simplifies deployment and versioning of model runs.

Batch Ingestion
Zürich OGD datasets update periodically rather than continuously; therefore, weekly batch ingestion is sufficient. The system fetches updates, processes them through Bronze → Silver, and triggers model recomputation when required.

Partitioning Strategy
To balance performance and manageability, we partition Silver data by month (after initially partitioning by day). This reduces the number of files while keeping partitions small enough for efficient parallel processing and reprocessing. The selected approach

avoids very small Parquet files (metadata overhead and “small files” inefficiency) and avoids excessively large files (reduced parallelism and slower retries). In our current setup, monthly partitions typically result in files around 15–20 MB.

Interoperability with the Data Warehouse
Only curated Gold outputs are loaded into the data warehouse. The warehouse then powers the marketing agency dashboards (visibility search interface) and real estate owner dashboards (portfolio revenue assessment. Heavy modelling logic remains in the data lake; the warehouse is optimized for querying, joining, and consumption.

3. DATA INGESTION

3.1. Data Source Documentation

The data ingestion layer consolidates multiple datasets published by the City of Zürich via its Open Government Data (OGD) platform and the Swiss geospatial hub GeoCat. Each dataset supports a specific modelling task within the data lake and is classified as either **dynamic**, meaning it is updated regularly and requires automated ingestion, or **static**, meaning it changes infrequently but provides essential spatial context.
Table 2: Data Sources summarizes the datasets used, their update frequency, access method, and their relevance for the facade advertisement use case. Together, these sources capture the full spectrum of urban mobility; motorized traffic, active mobility, and public transport, and provide the foundation required to compute potential impressions per facade at city scale.

Dynamic sources; motorized traffic, pedestrian/bicycle counts, and VBZ passenger volumes, change continuously and therefore require scheduled batch ingestion to keep the lake up to date. Facade geometries, in contrast, are static and are ingested during initial deployment or when new releases become available. All dynamic datasets accessed via the Zürich CKAN API share common access characteristics: no authentication is required, rate limits are mild, dataset metadata is retrieved via JSON endpoints, and pagination is either simple or unnecessary due to moderate file sizes. This uniformity enables a standardized ingestion framework across datasets.

No single dataset provides a complete representation of urban mobility. Only by combining all four sources can we derive hourly, multi-modal exposure estimates and produce city-wide estimations on facade impressions which support our use cases.

3.2. Pipeline Design

Data ingestion follows a fully automated **weekly batch workflow**, orchestrated via Amazon EventBridge. *Figure 1* illustrates how ingestion jobs populate the Bronze layer, forming the entry point into the data lake.

Each week, EventBridge triggers three independent Lambda-based ingestion pipelines corresponding to motorized traffic, pedestrian and bicycle traffic, and public transport (VBZ). All pipelines share a common ingestion framework, ensuring consistent logging, watermark management, and audit tracking across datasets. For CKAN-based datasets (motorized traffic and pedestrian/bicycle counts), the ingestion process retrieves dataset metadata, compares the most recent resource timestamp against a persisted state in S3, and downloads new data only if changes are detected. Raw CSV files are stored unmodified in the Bronze layer, after which they are normalized (column names standardized, timestamps converted, and data types enforced) and written as Parquet files to the Silver layer, partitioned by year and month.
VBZ passenger data follows a similar pattern but requires additional handling because it is distributed across multiple interlinked files (fact and reference tables). A single ingestion job retrieves all relevant files, computes file-level hashes, and compares them against stored hashes to detect updates. Only changed files trigger a rewrite of the corresponding Bronze and Silver partitions, ensuring efficient updates while preserving consistency between fact and reference data.

Table 4: Ingestion Technologies

COMPONENT	PURPOSE
AWS LAMBDA	Serverless ingestion for all datasets
AMAZON EVENTBRIDGE	Scheduled triggers for weekly batch run
AMAZON S3	Storage of Bronze and Silver outputs; storing watermarks and run state
DYNAMODB AUDIT TABLE	Tracks ingestion runs, errors, and results
CLOUDWATCH LOGS	Logging for traceability and debugging
GLUE DATA CATALOG	Schema registry for datasets

This architecture is lightweight, scalable, and cost-efficient, and is well suited for periodic ingestion of moderately sized public datasets.

3.3. Data Capture & Handling

Reliable ingestion is essential to ensure that downstream modelling and analytics remain reproducible and trustworthy. Because the source datasets vary in structure, update frequency, and stability, the ingestion layer incorporates mechanisms to handle schema evolution, transient failures, and data corrections.

Schema evolution is addressed through a two-stage strategy. In the Bronze layer, data is stored using a schema-on-read approach, preserving raw files exactly as delivered by the source. This guarantees full traceability and prevents information loss when upstream schemas change. In the Silver layer, data is standardized by enforcing consistent naming conventions, timestamp formats, and data types. Newly introduced columns are automatically integrated, allowing the data model to evolve incrementally without breaking existing pipelines.

To handle external dependencies such as the CKAN API, ingestion jobs implement retries for transient errors and degrade gracefully when no new data is available. All ingestion runs (successful, skipped, or failed) are recorded through structured CloudWatch logs and a shared DynamoDB audit table. This audit trail captures dataset identifiers, processing timestamps, watermark or hash values, and execution status, enabling transparent monitoring and efficient debugging.

Several OGD datasets occasionally correct historical values. To detect such changes, ingestion relies on file-based watermarks: timestamps for CKAN resources and cryptographic hashes for downloadable VBZ files. When a change is detected, the affected Bronze partitions are overwritten and only the corresponding Silver partitions are regenerated. This targeted overwrite strategy ensures that late-arriving or corrected data is incorporated cleanly, prevents duplicate records, and minimizes unnecessary recomputation.

4. DATA STORAGE

The data storage layer connects the data lake with the analytical and visualization components of the system. Its purpose is to persist curated, business-ready outputs in a form that supports efficient querying, spatial joins, and building/portfolio-level aggregation. Because our use case relies on geospatial intersections (facade visibility areas × mobility edges), storage decisions are driven by both analytical performance and spatial query capabilities.

4.1. Storage Layers and Services

We use a hybrid storage approach: **Amazon S3** for scalable data lake storage and **PostgreSQL with PostGIS** as an analytical serving layer.
S3 hosts all datasets across the medallion architecture: raw, immutable files in Bronze; standardized datasets in Silver; and curated, modelled outputs in Gold. This design leverages S3's scalability and low cost while maintaining flexibility for both tabular and geospatial formats (Parquet, GeoParquet, GeoPackage) without schema constraints.
For interactive analytics and dashboard consumption we use PostgreSQL with the PostGIS extension. This choice is motivated by native geospatial types, spatial indexing, and robust support for spatial joins, which are central to mapping mobility flows to facade visibility areas. In addition, the relational model enables consistent aggregation at facade and building level (e.g., EGID roll-ups) and integrates smoothly with BI tooling. Only validated and curated **Gold outputs** are loaded into the database so that end users query a consistent, business-ready dataset.

4.2. Data Formats and Optimization

We select storage formats based on performance and interoperability across the pipeline. CSV is used only as a source format and is retained in the Bronze layer as a faithful raw backup. Parquet is the standard format for Silver and Gold tabular datasets because its columnar compression supports efficient scans, time-series filtering, and repeated model execution on large mobility tables. For spatial datasets (facade visibility polygons and georeferenced flows), we use GeoParquet and GeoPackage, which preserve spatial metadata while remaining compatible with geospatial libraries and database loaders.

On S3, datasets are organized to minimize scan cost and simplify incremental processing. Data is structured by domain (e.g., traffic, pedestrian, VBZ, facades) and partitioned by year and month (or date where applicable). This allows models and analytics to load only relevant subsets rather than entire datasets.

4.3. Database Storage Architecture and Implementation

The analytical database is designed to support two main workloads: (1) efficient spatial joins between facade visibility geometries and mobility edges and (2) fast retrieval of pre-aggregated impression metrics for dashboard interaction. The database runs PostgreSQL with PostGIS on AWS RDS.

The schema follows a clear separation between **geometry reference tables** and **time-dependent flow tables** (*Figure 4*). Geometry tables store facades and mobility networks (edges), while flow tables store hourly metrics keyed by composite identifiers (e.g., edge_id and hour). This design avoids duplicating geometry in time-series fact tables and supports consistent joins across modes. Public transport

data is stored as georeferenced segments with the corresponding hourly passenger-related metric, enabling it to be treated consistently with other mobility modes.
Spatial relationships between facades and mobility networks are derived through spatial joins (e.g., intersection tests). To make these joins performant, geometry columns are indexed using spatial indexing. For dashboard usage, expensive spatial joins and aggregations are not executed on every user interaction; instead, the system caches the final result in a materialized view that stores one row per facade with impressions metrics (total and per mode). This caching strategy ensures interactive performance while keeping the modelling logic in the lake and the consumption logic in the warehouse.
To support accurate spatial computation, geometry is stored and processed in the Swiss coordinate system (EPSG:2056). For visualization use cases, facade centroids can additionally be exposed in WGS84 for easy integration with web-based mapping.

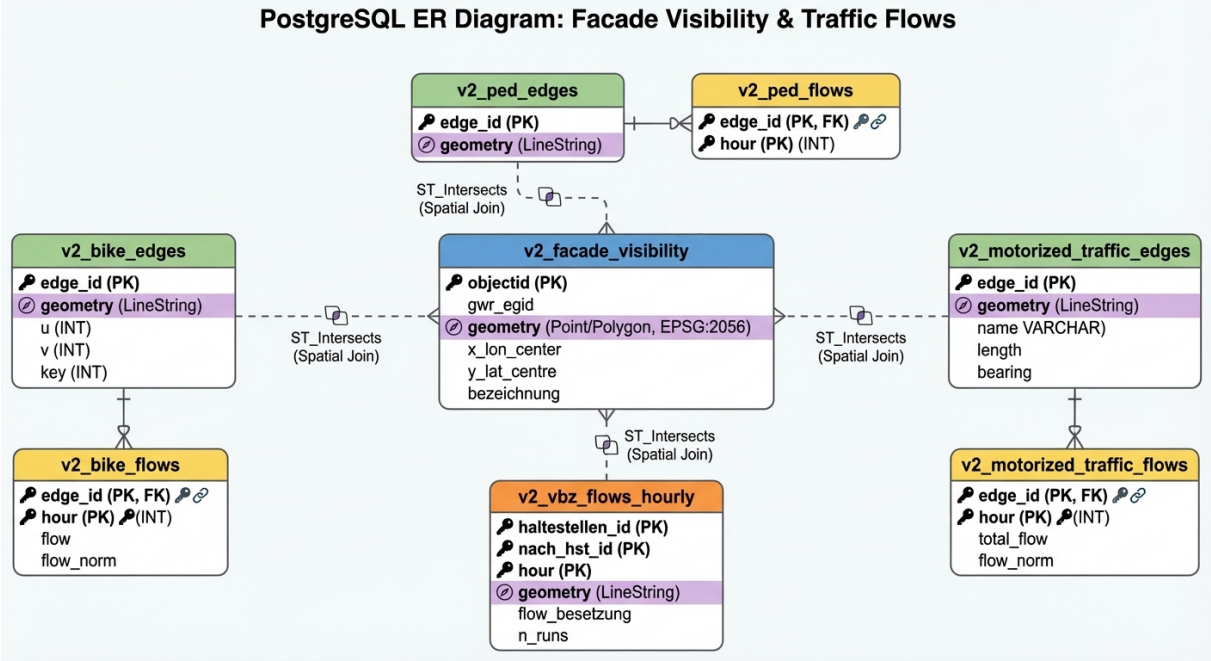


Figure 2: ER Diagram of PostgreSQL Database in the Data Warehouse

5. DATA TRANSFORMATION

After ingestion, all datasets enter the Bronze layer of the medallion-based data lake in their raw, source-shaped form. This design preserves a complete historical record and allows incremental reprocessing as models evolve. From this point onward, the transformation layer converts raw measurements into structured, spatially consistent, and analytically meaningful datasets that ultimately power facade visibility estimation and downstream analytics.

Figure 3 illustrates the transformation pipeline within the medallion architecture. Conceptually, the transformation logic is divided into two stages with clearly separated responsibilities: **uniform normalization and format standardization**, followed by **model-specific geospatial transformations**. This separation decouples lightweight, schema-oriented processing from computationally intensive spatial modelling and simplifies both maintenance and scaling.

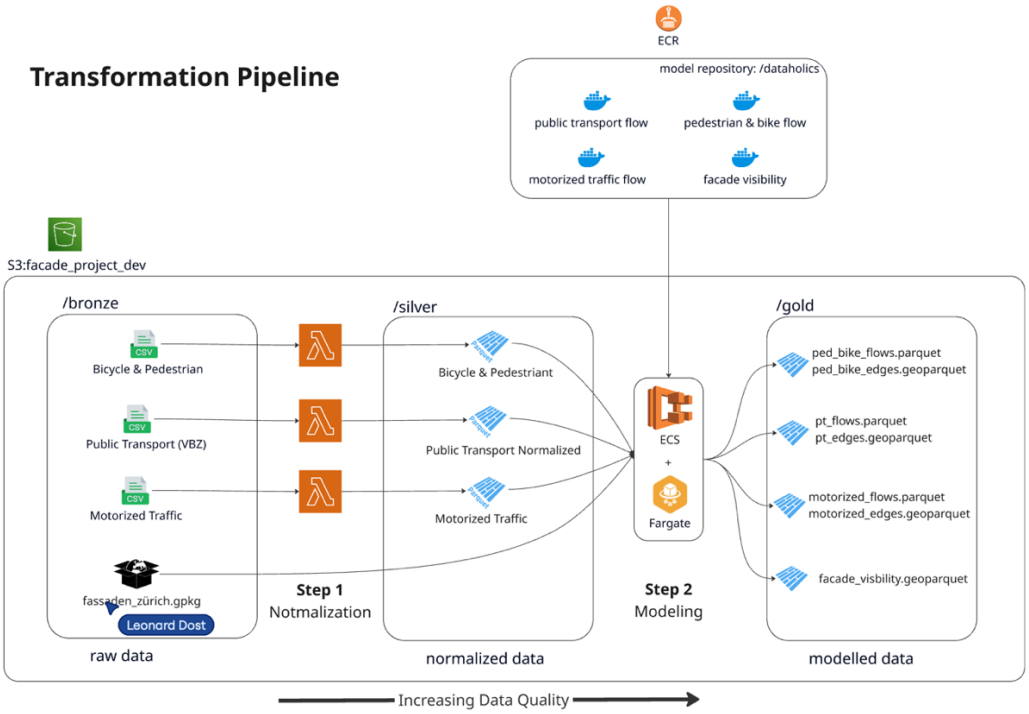


Figure 3: Transformation Pipeline and Medallion Architecture.

5.1. Normalization and Format Standardization

The first transformation stage operates directly on Bronze-layer datasets and serves to harmonize heterogeneous source files into a consistent, model-ready representation. At this stage, transformations are deliberately limited to structural changes: column names are normalized to snake_case, timestamps are converted to consistent formats, and raw CSV inputs are converted to Parquet. No semantic interpretation, aggregation, or filtering is applied. Normalization is executed as part of the ingestion process itself, within the same AWS Lambda functions that retrieve the source data. As described in Chapter 3, separate ingestion Lambdas handle motorized traffic, pedestrian and bicycle traffic, and public transport data, each writing standardized outputs to the Silver layer. By restricting Silver-layer transformations to structural alignment only, the data remains close to its original meaning while being optimized for analytical access and interoperability across the data lake.

5.2. Geospatial Modelling in ECS

The core analytical value of the system is created in the second transformation stage, where standardized Silver datasets are transformed into spatially explicit, business-ready outputs. This stage consists of multiple independent geospatial models that convert point-based or segment-based mobility measurements into continuous flow estimates and facade visibility areas. These transformations involve spatial buffering for facade visibility area construction, network-based flow propagation for different mobility modes and temporal alignment of hourly flows. Due to their computational complexity and dependency on geospatial libraries in python that depend on low level libraries such as GDAL (which is not available on Lambda Layers) they are executed in AWS ECS using Fargate rather than AWS Lambda.

Each model is implemented as a containerized Python application, developed locally and deployed via Docker images stored in AWS ECR. Executing the models in ECS provides a consistent runtime environment, reproducible execution, and isolated parallel processing of independent pipelines. All models consume standardized Silver inputs and write their results exclusively to the Gold layer, where they become production-ready assets. Although the models share a common execution infrastructure, they are logically independent and can be versioned, re-run, or extended without affecting other pipelines.

5.2.1. Facade Visibility Area Model

The facade visibility model translates facade geometries into spatial representations of potential visual exposure. Its purpose is not to predict whether a facade is seen at a specific moment, but to define the area from which a facade could plausibly be perceived by people moving through the city. These visibility areas form the spatial basis for combining facades with modelled mobility flows. Starting from facade line geometries, the model projects a visibility region outward based on facade orientation. The region is defined by a configurable distance and angular opening, forming a cone-like shape that approximates the facade's field of view into public space. Applying this logic consistently across all facades produces a complete set of visibility polygons. Facades can be aggregated by building identifier (EGID), enabling cumulative visibility estimation at building level. *Figure 4* illustrates example visibility polygons generated by the model.

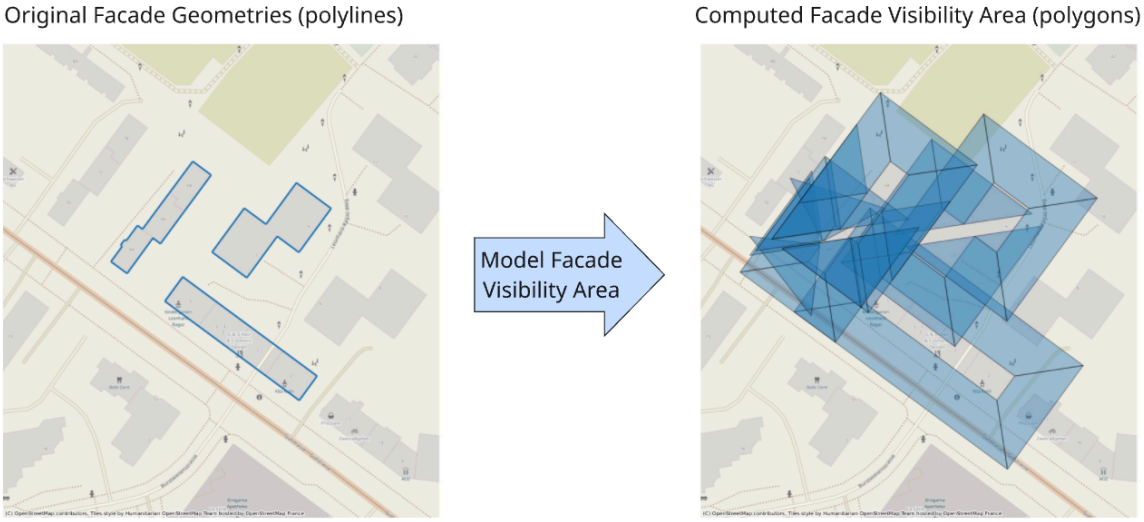


Figure 4: Computed facade visibility polygons illustrating the spatial regions from which facades may be visually exposed.

5.2.2. Public Transport Flow Model

The public transport flow model transforms VBZ passenger data into georeferenced flows along the public transport network. While the raw VBZ dataset provides passenger volumes between stops and departure times, it lacks spatial geometries suitable for spatial analysis. The goal of this model is therefore to convert stop-to-stop passenger counts into spatial line segments that can be intersected with facade visibility areas.

The model begins with the VBZ “Reisende” fact table and enriches it using reference data such as stop metadata and day types. Passenger volumes are aggregated to flows between consecutive stops, either on an hourly basis or as daily totals, depending on the configured output resolution. This flexibility allows analysis of public transport exposure at different temporal granularities. To derive geometries, the model combines VBZ stop data with the ZVV GTFS feed. Stop names are normalized to improve matching, and GTFS coordinates are used to georeference VBZ stops. The model then retrieves the drivable street network for Zürich from OpenStreetMap, snaps each stop to the nearest network node, and computes shortest paths between stop pairs. These paths are converted into LineString geometries, producing realistic, street-following representations of public transport routes. *Figure 5* shows an example visualization of aggregated public transport flows for a typical workday.

The model writes four outputs to the Gold layer: two full georeferenced datasets (GeoPackage and GeoParquet) for GIS usage and scalable analytics, and a normalized split consisting of an edges-only GeoParquet and a geometry-free flows table in Parquet format. This normalized structure mirrors the outputs of other mobility models and supports efficient loading into the downstream data warehouse.

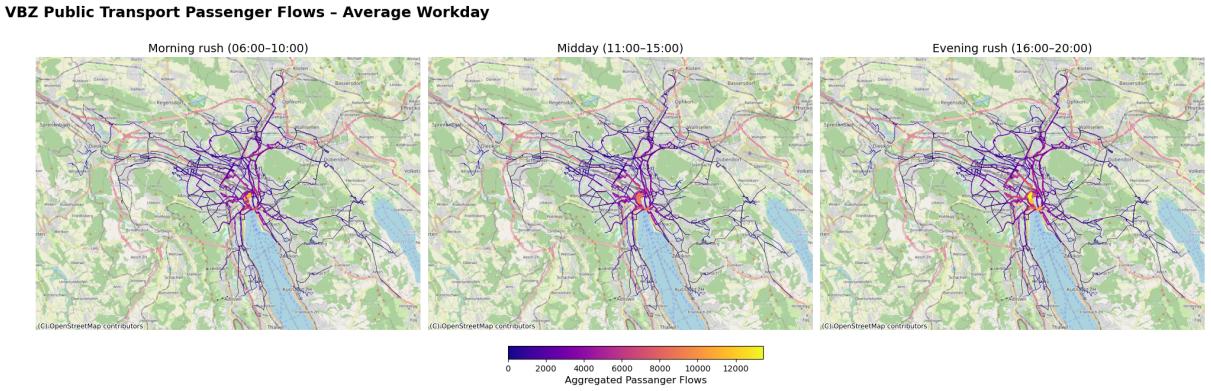


Figure 5: Aggregated passenger flows derived from VBZ data and mapped to georeferenced network segments. The visualization illustrates temporal differences between morning rush, midday, and evening rush periods.

5.2.3. Motorized Traffic Flow Model

The motorized traffic flow model converts counting-station measurements from the City of Zürich into a city-wide estimate of vehicle flows that can be intersected with facade visibility areas. Since the raw data only provides observations at discrete stations, the model propagates these measurements onto a drivable street network.

Input files are loaded from partitioned Parquet data and aggregated into a **typical day profile** by computing mean traffic volume per hour of day for each station and direction. Each station-direction seed is assigned a propagation direction using simple heuristics: “inbound/outbound” labels are interpreted relative to the city center, while common free-text direction labels are mapped via a curated dictionary. The model then retrieves

the OpenStreetMap street network for Zürich, snaps stations to the nearest road segment, and propagates flows along the network with distance decay and directional weighting so that flows follow plausible travel directions.

To prevent unrealistically inflated values, the propagation includes **mass-balance safeguards**. Locally, the flow leaving a node is distributed across eligible outgoing streets as shares that sum to one, ensuring that flow is redistributed rather than multiplied. In addition, a global correction step ensures that the total propagated flow per hour remains consistent with the total measured input flow for that hour. Propagation is limited by maximum distance and stops when flows decay below a minimum threshold, which keeps the results stable and computationally efficient.
The model writes outputs both in a full georeferenced form for inspection and GIS use and in a normalized form that separates edge geometries from hourly flow values. A stable edge identifier is used across outputs to support efficient downstream joins and warehouse loading. *Figure 6* shows the modelled motorized traffic flows across the City of Zürich.

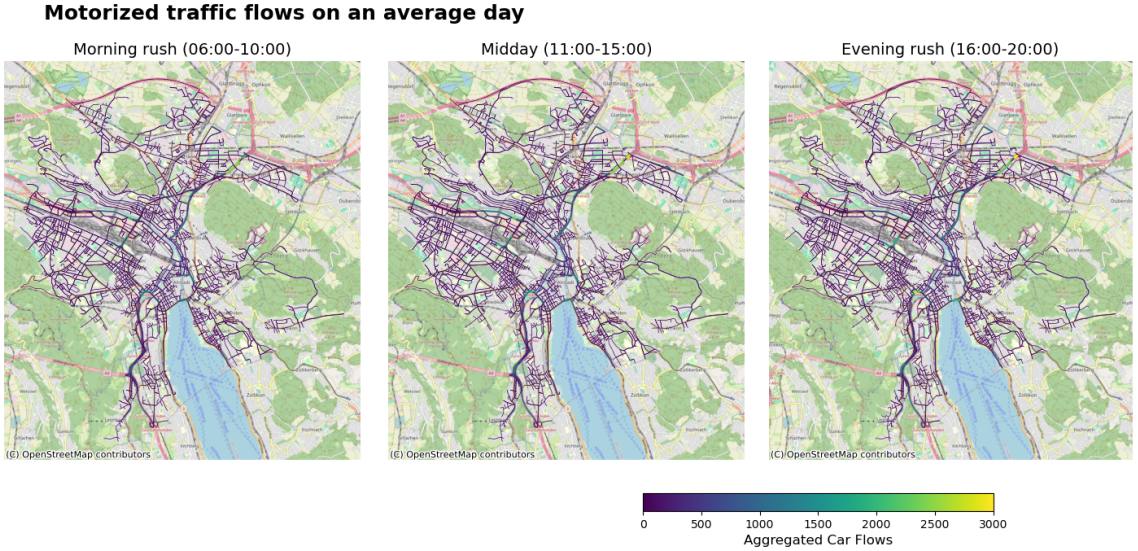


Figure 6: Aggregated motorized traffic flows mapped to the street network for different time windows, illustrating temporal variations and concentration along major corridors.

5.2.4. Pedestrian and Bicycle Flow Model

The pedestrian and bicycle flow model follows the same core propagation logic as the motorized traffic flow model, adapting it to non-motorized mobility patterns. The input data consists of counting-station measurements for pedestrians and cyclists provided by the City of Zürich, which are likewise spatially sparse and therefore require network-based propagation.

As with motorized traffic, observations are transformed into propagation seeds and mapped onto OpenStreetMap networks. Separate networks are used for each mode; a walking network for pedestrians and a cycling network for bicycles, to ensure that flows are propagated only along paths that are realistically accessible. Directional information is derived where available, but is applied more softly than for motorized traffic, reflecting the less constrained movement behavior of pedestrians and cyclists.
Flow propagation reuses the same **mass-balanced approach** described in Section 5.2.3. At each step, incoming flow is redistributed across eligible outgoing edges without amplification, combined with distance-based decay and a global hourly mass-balance correction. This ensures that total pedestrian and bicycle flows remain stable

and comparable to the original measurements while still producing spatially continuous flow estimates.

Mode-specific parameters account for shorter typical travel distances and local movement patterns, resulting in more compact propagation ranges than for motorized traffic. The outputs are written in a fully normalized form, separating edge geometries from time-dependent flow tables. This structure avoids geometry duplication, remains consistent across all mobility models, and allows pedestrian, bicycle, and motorized flows to be combined seamlessly in downstream spatial analysis and facade visibility estimation.

5.3. Data Cleaning & Preparation

Data cleaning and preparation are embedded throughout the transformation pipeline rather than implemented as a single isolated step. Structural cleaning, such as column normalization, type enforcement, and timestamp alignment, is handled systematically during the Bronze-to-Silver transformation. More advanced preparation occurs within the modelling pipelines themselves, including spatial filtering, geometry validation, temporal alignment, and enrichment with network context. Keeping preparation logic close to its analytical context avoids duplication and improves transparency.

5.4. Data Quality & Validation

Data quality is ensured through a combination of architectural safeguards and integrated validation practices. The medallion architecture establishes clear quality boundaries between raw, standardized, and modelled data. Ingestion watermarks prevent silent data loss, schema normalization surfaces unexpected structural changes, and model execution logs provide traceability. Spatial modelling steps inherently expose anomalies such as invalid geometries or disconnected network components. This approach favors early detection and traceability over rigid centralized checks, which are often impractical for heterogeneous geospatial data.

5.5. Scalability & Reusability

Scalability and reusability are core design goals of the transformation layer. Separating normalization from modelling and executing computationally intensive tasks in ECS Fargate allows the system to scale with data volume and model complexity. Each model is implemented as an independent, containerized pipeline that can be developed, tested, re-executed on historical data, and evolved without affecting other components. The consistent structure of Silver and Gold outputs further supports reuse, enabling new models to be integrated without changes to ingestion or normalization logic.

6. DATA WAREHOUSE

This chapter describes how the modeled datasets produced in the data lake are consolidated into a data warehouse and made accessible for analytical use. In contrast to traditional architectures where substantial transformation logic resides inside the warehouse, our approach deliberately shifts normalization and modelling upstream into the data lake. As a result, the data warehouse primarily serves as an efficient, query-oriented layer that exposes curated data for analysis and visualization.

6.1. Business Goals and Objectives

The data warehouse is designed to support two complementary business perspectives within the facade advertising ecosystem. From the demand side, marketing agencies require a reliable way to identify facade surfaces with high advertising potential. Their core interest lies in estimating visual impressions generated by different mobility modes (pedestrian, bicycle, motorized traffic, and public transport) across time. The warehouse provides structured, queryable access to facade-level exposure metrics, enabling agencies to filter, rank, and compare locations based on expected visibility. This supports more targeted campaign planning and expands the set of viable advertising surfaces beyond intuition-driven selection.

From the supply side, real estate owners and portfolio managers require transparency regarding the economic viability of facade advertising. The warehouse supports this perspective by aggregating facade-level exposure metrics to building and portfolio level and linking them to monetization benchmarks. This enables owners to compare assets, assess revenue potential, and evaluate expected returns in a consistent and data-driven manner.

These two perspectives converge on a shared analytical foundation. By anchoring all metrics at facade level and enabling roll-ups to buildings and portfolios using EGID as the central building identifier, the warehouse ensures that insights remain coherent across analytical scales. This shared structure improves comparability, supports price discovery, and increases trust in the derived metrics.

6.2. Data Warehouse Modeling

The warehouse follows a star-oriented analytical design adapted to spatial data, as outlined in *Chapter 4*. Quantitative measures such as mobility flows and facade-level exposure metrics are stored in fact tables, while descriptive attributes (facades, buildings, time, and mobility mode) are represented through dimension tables.

Spatial relationships are a defining characteristic of the model. Rather than relying solely on relational keys, flow edges are linked to facades through spatial joins, enabling exposure metrics to be derived from geometric intersection. The use of PostGIS allows these spatial relationships to be expressed directly within the warehouse while remaining performant through spatial indexing. Natural identifiers such as facade IDs and EGID are preserved for analytical clarity, while surrogate keys are used internally where appropriate for efficiency.

6.3. Architecture

The data warehouse is built directly on top of the **gold layer of the data lake**, which contains fully processed and normalized datasets. These datasets are stored as

Parquet and GeoParquet files and represent the authoritative analytical inputs for the warehouse.

Once loaded into the database, the warehouse provides a unified relational and spatial view across all modeled domains. From there, the data is consumed by a visualization layer implemented in Tableau, which connects directly to the database to power interactive dashboards and reports. This separation ensures that heavy computation and modeling remain outside the warehouse, while the warehouse itself remains optimized for fast querying and aggregation.

6.4. Data Preparation

The suitability of the data sources stems from the fact that all core modeling steps (flow propagation, normalization, spatial alignment, and temporal structuring) are completed before data reaches the warehouse. The gold-layer datasets are therefore already close to a relational normal form, minimizing the need for complex transformations during loading.

Data extraction consists of loading curated Parquet and GeoParquet files from the gold layer into the PostGIS database. Transformation steps inside the warehouse are intentionally minimal and focus on lightweight tasks such as enforcing schemas, creating indexes, and materializing commonly used aggregates where necessary. Spatial enrichment, such as linking flow edges to facades, is performed using PostGIS functions.

The loading strategy primarily follows **incremental loads**, allowing new time periods or updated model runs to be appended without reprocessing the full dataset. Full reloads remain possible for reproducibility or major model changes, but are not required for routine operation.

Overall, this ELT-light approach ensures a clear separation of concerns: complex modeling logic is handled in the data lake, while the data warehouse remains a performant, stable foundation for analytical queries and visualization.

7. DATA VISUALIZATION

7.1. Purpose and Role of the Visualization Layer

The data visualization layer translates the modeled, mobility-driven exposure metrics into actionable insights for decision-makers. In alignment with the two use cases introduced in *Chapter 1*, the visualization is implemented as **two complementary dashboards**, each tailored to a specific user group while relying on the same analytical foundation.

The first dashboard addresses the demand-side perspective and supports marketing agencies in identifying facade surfaces with high advertising potential. The second dashboard targets the supply-side perspective, enabling real estate owners and portfolio managers to assess the revenue potential of their buildings or portfolios. Both dashboards are powered by the same data warehouse, which ingests curated outputs from the Gold layer of the data lake. This ensures analytical consistency across use cases while allowing role-specific interpretation and interaction. Screenshots of both dashboards are provided in the appendix. This chapter focuses on the conceptual design, analytical intent, and business relevance of the visualization layer.

7.2. Design Principles and Analytical Approach

The visualization design follows established principles from business intelligence and visual analytics. It adopts a **task-oriented structure**, where each element directly supports a concrete decision task such as filtering, ranking, comparison, or evaluation of advertising locations. Rather than encouraging open-ended exploration, the dashboards are designed to guide users toward informed decisions. A second guiding principle is **progressive disclosure**. High-level key metrics are presented first, allowing users to quickly assess overall performance, while more detailed spatial and contextual information becomes available through interaction. This reduces cognitive load and helps users focus on relevant insights without being overwhelmed by technical complexity.

Quantitative exposure metrics are presented in aggregated and tabular form to support precise comparison across facades and buildings. Spatial context is provided through map views and street-level imagery, enabling users to validate numerical rankings against the physical environment. This combination of abstract metrics and real-world context increases trust in the results and supports practical decision-making. Interactivity is used selectively to support **scenario-based reasoning**. Users can adjust weighting assumptions or economic parameters and immediately observe how rankings or revenue estimates change, without requiring technical knowledge of the underlying models.

7.3. Dashboard 1: Facade Visibility for Marketing Agencies

The first dashboard supports the demand-side use case and is designed for media planners, campaign managers, and marketing analysts. Its primary objective is to identify and compare facade surfaces based on their potential visual reach.
Exposure metrics are presented at facade and building level, to reflect how advertising locations are typically evaluated and marketed. The dashboard enables users to rank

buildings by total impressions and analyze how exposure is distributed across mobility modes such as pedestrians, cyclists, motorized traffic, and public transport.

A key feature is the ability to dynamically adjust **mode-specific weighting factors**. This allows agencies to tailor the analysis to specific campaign goals, for example prioritizing pedestrian exposure for street-level branding or public transport exposure for commuter-focused campaigns. Rankings and totals update immediately, supporting rapid comparison across alternative scenarios.

To complement quantitative metrics, the dashboard integrates spatial and visual context. Map views show the location of selected buildings, while street-level imagery allows users to assess facade orientation, surrounding streetscape, and potential obstructions. This integration supports end-to-end decision-making within a single interface, from data-driven ranking to qualitative feasibility assessment.

7.4. Dashboard 2: Revenue Potential for Real Estate Owners

The second dashboard addresses the supply-side use case and reframes the same exposure data from an economic perspective. It is intended for real estate owners and portfolio managers who want to assess whether facade advertising represents a viable revenue opportunity for their assets.

Users can select individual buildings or entire portfolios and view aggregated impression metrics derived from the same warehouse tables used by the marketing dashboard. To translate visibility into economic terms, the dashboard introduces a configurable **cost-per-mille (CPM)** parameter. Based on this input, the system derives indicative revenue estimates at daily, weekly, and monthly level.

This approach allows owners to explore how changes in pricing assumptions or portfolio composition affect potential income, without embedding complex financial models into the visualization layer. The resulting KPIs provide a transparent, first-order estimate of monetization potential that can support strategic discussions and investment decisions.

As in the marketing dashboard, spatial context and street-level imagery are included to support practical evaluation. This enables users to combine quantitative revenue estimates with qualitative assessment of facade characteristics and urban context.

7.5. Shared Data Backbone and Consistency Across Dashboards

Both dashboards are powered by the **same data warehouse** and rely exclusively on curated datasets produced in the Gold layer of the data lake. All computationally intensive steps; including flow propagation, spatial joins, normalization, and aggregation, are completed upstream. As a result, the visualization layer remains lightweight and focused on interpretation rather than data processing.

By exposing the same analytical foundation through two role-specific dashboards, the system connects demand and supply perspectives within a single platform. Marketing agencies gain transparency into visibility potential, while property owners gain insight

into monetization opportunities. This shared data backbone ensures comparability, reduces information asymmetry, and demonstrates how the data warehouse directly enables business value across stakeholders.

Notes on Scope and Evaluation

The dashboards are implemented as functional prototypes that demonstrate the analytical capabilities of the system. While the current focus is on the demand-side and supply-side core use cases, the visualization architecture is designed to be extensible. Additional metrics, user roles, or analytical views can be integrated without changes to the underlying data lake or warehouse structure.

Links to the Dashboards

Facade Impressions (Dashboard 1): https://public.tableau.com/app/profile/leonard.dost/viz/FacADeDashbaord/FacADeDashboard

Facade Revenue (Dashboard 2): https://public.tableau.com/app/profile/leonard.dost/viz/FacADeRevenueDashbaord/FacADeRevenueDashboard

8. CONCLUSIONS

8.1. Discussion of the Solution

The FacADe project demonstrates that it is technically feasible to combine heterogeneous mobility data sources with geospatial modeling to estimate facade-level advertising exposure at city scale. By shifting complex transformations and spatial modeling into the data lake and using the data warehouse primarily as a serving layer, the system achieves a clear separation of concerns. This architectural choice proved effective in keeping the warehouse performant while maintaining analytical flexibility.
A major strength of the solution is the **end-to-end consistency** of the pipeline. All modeled flows, facade visibility geometries, and aggregated exposure metrics are generated upstream, versioned in the data lake, and exposed in a unified form through the data warehouse. This enables both dashboards to rely on the same data foundation while supporting different business perspectives. The use of PostGIS for spatial joins and aggregation proved to be a suitable trade-off between expressiveness and performance for this type of analysis.

However, several trade-offs and limitations became apparent. The flow propagation models necessarily rely on simplifying assumptions about movement behavior, directionality, and decay with distance. While safeguards such as mass-balance constraints prevent unrealistic blow-ups, the resulting flows remain estimates rather than ground truth. In addition, the dependence on OpenStreetMap network data introduces sensitivity to network completeness.

From a technology perspective, the AWS-based architecture (S3, ECS, RDS, Tableau) worked reliably and scaled well within the project scope. At the same time, this setup introduces operational complexity and cost considerations that would need to be evaluated carefully in a real commercial deployment.

8.2. Project Outcomes

The primary outcome of the project is a **fully functional analytical prototype** that spans data ingestion, transformation, storage, and visualization. The complete infrastructure was implemented successfully and operated reliably during testing.
The final dashboards demonstrate that the system can:

- identify and rank facades and buildings based on modeled exposure,
- distinguish between different mobility modes and temporal patterns,
- and translate exposure metrics into indicative revenue estimates using configurable pricing assumptions.

A key limitation emerged in the **availability and spatial coverage of pedestrian and bicycle data**. The City of Zürich provides only a small number of pedestrian and cyclist counting stations compared to motorized traffic sensors. As a result, flow propagation for these modes is spatially sparse and highly uncertain in many parts of the city. This significantly reduces the reliability of pedestrian and bicycle exposure estimates outside well-covered areas.

This limitation is particularly critical from a business perspective. Pedestrian exposure is often the most valuable form of facade advertising, as pedestrians remain within the

visual field longer and are more likely to engage with physical advertisements. The lack of dense pedestrian and cyclist data therefore poses a direct risk to the overall value proposition of the system. While the motorized and public transport components perform robustly, the missingness in slow-mobility data limits the completeness of the exposure model.

8.3. Future Work

Several directions for future work emerged clearly from the project.
On the modeling side, **probabilistic approaches** could significantly improve robustness. Instead of producing a single deterministic flow estimate, Monte Carlo–based propagation could simulate multiple plausible realizations of traffic flows from the same input data. Aggregating these simulations would allow uncertainty to be quantified explicitly and reduce sensitivity to individual assumptions or sparse measurements.

Another promising extension is the integration of **computer vision on facade imagery**. Using deep learning models on street-level images could help estimate the actual usable advertising surface on facades, accounting for windows, architectural elements, and obstructions. This would allow exposure metrics to be weighted by effective advertising area rather than treating all facades as homogeneous surfaces.
From a business and data perspective, the system would benefit strongly from real-world usage. Once both marketing agencies and property owners interact with the platform, it would become possible to collect empirical data on achieved prices, conversion rates, and campaign performance. Over time, this could enable the construction of a pricing database that reflects actual supply and demand for facade advertising in Zürich, allowing revenue estimates to be grounded in observed market behavior rather than assumed cost-per-mille (CPM) values.

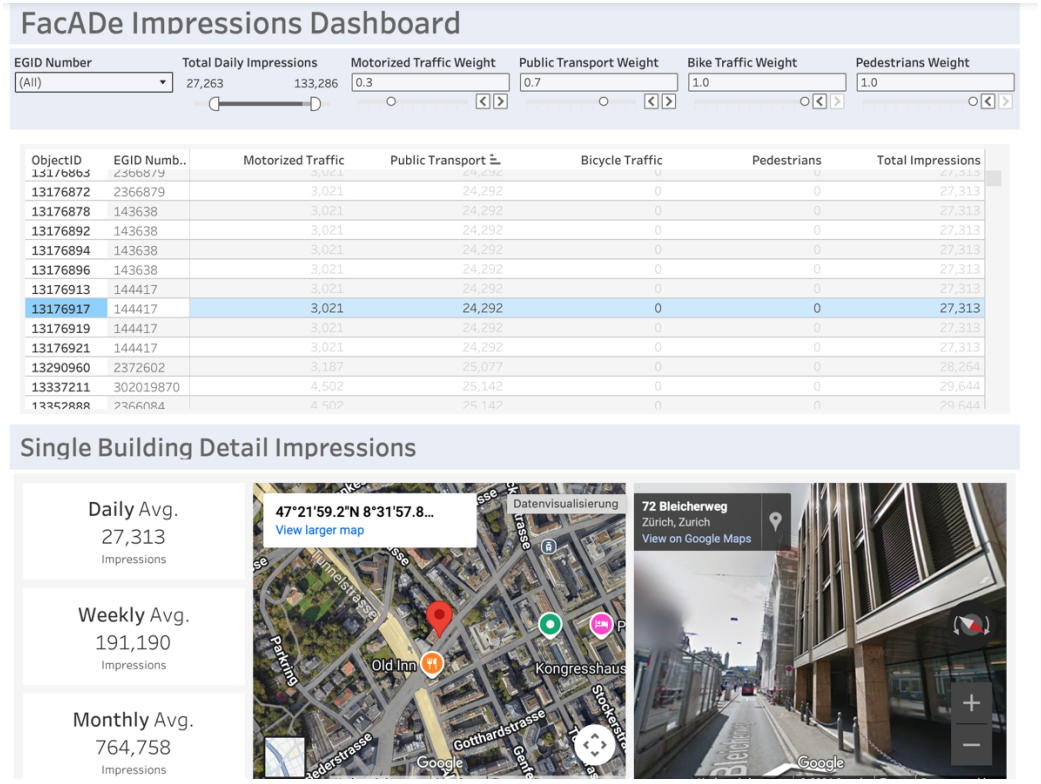
8.4. GenAI Declaration

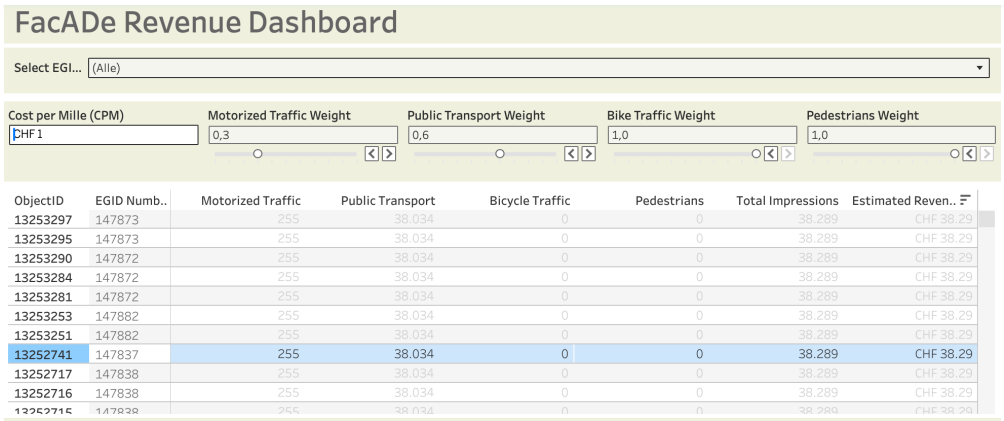
Generative AI tools were used as a supporting instrument during the project, primarily to discuss architectural and implementation decisions in AWS, support in coding projects and to refine and edit the written report. All core system design decisions, modeling logic, implementation, and analytical reasoning were developed and validated by the project team. Generative AI was not used to generate data, implement production code without review, or replace critical analytical judgment.

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APPENDIX A: Dashboards





APPENDIX B: Project Leads

Code Project Leads

Responsibilities	Amrein	Campell	Dost
facade-motorized-traffic-ingestion **(Ingestion, Clean & Partition)**	X		
facade-vbz-small-tables **(Ingestion, Clean & Partition)**	X		
facade-vbz-reisende-batch **(Ingestion, Clean & Partition)**	X		
facade-pedestrian-velo **(Ingestion, Clean & Partition)**	X		
facade-streetview **(Google Streetview API)**			X
facade-warehouse **(Relational Database System Ingestion)**			X
bike-peds-flow **(FacADe Model)**		X	
facade / run_visibility_flow **(FacADe Model)**		X	
traffic-flows **(FacADe Model)**		X	
Vbz-flows **(FacADe Model)**		X	
glue_loader_jdbc.py **(Relational Database System Ingestion)**			X

Topic Leads: Grade by Students

Evaluation Criteria	*Maximum possible points*	*Dataholics*		
		Amrein Marcel	*Campell Fadri*	*Dost Leonard*
1. Introduction	**5**	**5**	**5**	**5**
a) Topic introduction b) Motivation c) Problem definition 6 Goals d) Business/Research questions e) Busines intelligence questions		x	x	x
2. Data Lake	**5**	**5**	**5**	**5**
a) Architecture b) Data Assumptions		x	x	x
3. Data Ingestion (Data Lake)	**5**	**5**	**0**	**0**
a) Data Source Descriptions b) ETL Pipelines		x		
4. Data Storage	**5**	**5**	**5**	**5**
a) Databases and/or storage services		x	x	x
5. Data Transformation	**5**	**5**	**5**	**0**
a) Data Quality		x	x	
6. Data Warehouse	**5**	**0**	**5**	**5**
a) Architecture b) Data Preparation c) Data Warehouse Modeling			x	x
6. Data Visualization	**5**	**0**	**0**	**5**
a) Theoretical model b) Data Warehouse Visualization				x
8. Conlusions	**5**	**5**	**5**	**5**
a) Propse discussion of the solution b) Proper discussion of the project outcomes c) Future work		x	x	x
GRADE BY STUDENT / 40	**40**	**30**	**30**	**30**